# Current status of the DY absolute cross-section measurements study

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# Outline


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8. Latest Vs Existing Reconstruction Efficiencies (LD2)

# Events from runs 2 and 3 (Input Files Used)

**Note:** Currently using all the runs saved in runs 2 and 3.

**Harsha's ROOT files:**
/seaquest/users/harshaka/e906_project/e906-root-ana/work_gpvm/scripts/results_runFinalTree/merged_results_e906_mixing/:

- merged_RS57_LH2_1_1138.root
- merged_RS57_LD2_3_1138.root
- merged_RS57_Empty_2_1138.root
- merged_RS59_LH2_1_465.root
- merged_RS59_LD2_3_466.root
- merged_RS59_Empty_2_466.root
- merged_RS62_LH2_1_1234.root
- merged_RS62_LD2_3_1237.root
- merged_RS62_Empty_2_1234.root
- merged_RS70_LH2_1_264.root
- merged_RS70_LD2_3_266.root
- merged_RS70_Empty_2_267.root

**Abi's ROOT files:**
/seaquest/users/apun/e906_projects/rs67_merged_files/:

- merged_RS67_3089LH2.root
- merged_RS67_3089LD2.root
- merged_RS67_3089flask.root

# Event Selection Used: Chuck Cuts (DocDB 2111-V42) & Fiducial Cuts

**Chuck Cuts (**DocDB2111-V42**)** with beam_offset $= 1.6$ cm and mass cut $= 4.2$ GeV/$c^2$:

## Dimuon Cuts (dimuon_cut_2111_v42)

- Vertex: $|dx| < 0.25$, $|dy - 1.6| < 0.22$, $-280 < dz < -5$ cm
- Transverse Vertex: $dx^2 + (dy - 1.6)^2 < 0.06$ cm$^2$
- Momentum: $|dp_x| < 1.8$, $|dp_y| < 2.0$, $dp_x^2 + dp_y^2 < 5.0$ (GeV/$c$)$^2$, $38 < dp_z < 116$ GeV/$c$
- Kinematics: $4.2 <$ mass $< 8.8$ GeV/$c^2$, $-0.1 < x_F < 0.95$, $0.05 < x_T \leq 0.58$, $|\cos\theta| < 0.5$
- Fit & Separation: $|$trackSeparation$| < 270$ cm, $\chi^2_{\text{dimuon}} < 18$

## Single Track Cuts (track1_cut & track2_cut, $i = 1, 2$)

- Target/Dump $\chi^2$: $\chi^2_{i,\text{target}} < 15$, $\chi^2_{i,\text{target}} < 1.5\chi^2_{i,\text{upstream}}$, $\chi^2_{i,\text{target}} < 1.5\chi^2_{i,\text{dump}}$
- Track Quality: nHits$_i > 13$, $\chi^2_i/($nHits$_i - 5) < 12$, $-320 < z_{i,v} < -5$ cm
- Apertures: $x_{i,t}^2 + (y_{i,t} - 1.6)^2 < 320$, $16 < x_{i,d}^2 + (y_{i,d} - 1.6)^2 < 1100$ cm$^2$
- Station 1 & 3: $9 < p^{\text{st1}}_{z,i} < 75$, $|p^{\text{st1}}_{y,i}| > 0.02$ GeV/$c$, $y_{i,\text{st1}}/y_{i,\text{st3}} < 1$, $y_{i,\text{st1}} \cdot y_{i,\text{st3}} > 0$ -
- $\Delta p$ Matching: $||p^{\text{st1}}_{x,i} - p^{\text{st3}}_{x,i}|0.416| < 0.008$, $|p^{\text{st1}}_{y,i} - p^{\text{st3}}_{y,i}| < 0.008$,

## Track Pair Cuts (tracks_cut_2111_v42)

- $|\chi^2_{1,\text{target}} + \chi^2_{2,\text{target}} - \chi^2_{\text{dimuon}}| < 2.0$
- $y_{1,\text{st3}} \cdot y_{2,\text{st3}} < 0$ (opposite top/bottom halves, no beam offset)
- nHits$_1$ + nHits$_2 > 29$ and nHits1St1 + nHits2St1 $> 8$
- $|x_{1,\text{st1}} + x_{2,\text{st1}}| < 42$ cm

## Occupancy Cuts (occ_cut_2111_v42)

- $D_1 < 400$, $D_2 < 400$, $D_3 < 400$, and $D_1 + D_2 + D_3 < 1000$
- ***Note:*** **Intensity cuts (**intensity_cut_2111_v42**) are not applied.**

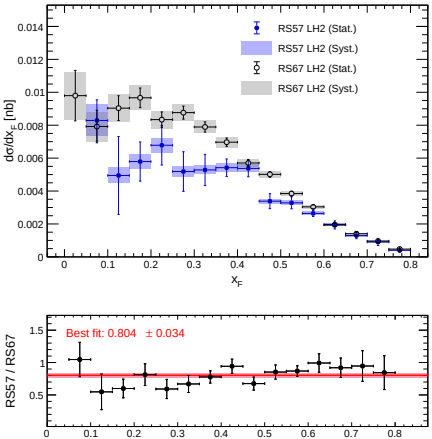
## Additional Fiducial Cut Applied

**In addition to the Chuck cuts above, we apply the kinematic fiducial cut:**

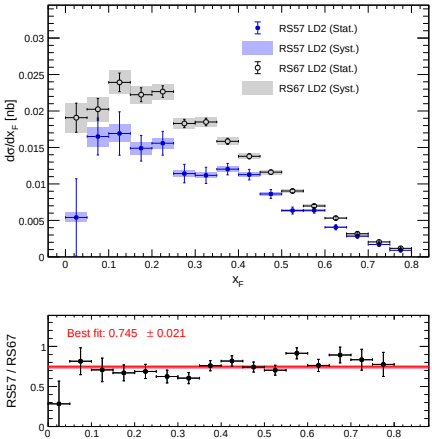
$(4.2 <$ mass $< 8.8$ GeV/$c^2$) & ($0.0 < x_F < 0.8$) & ($0 < p_T < 1.8$ GeV/$c$)

# Cross-Section Ratio RS57 to RS67 (xF bins)

**LH2 (RS57 vs RS67)**

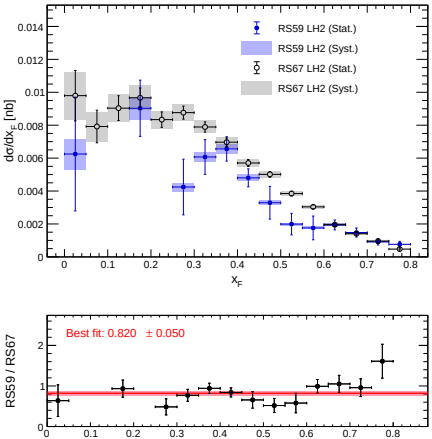



**LD2 (RS57 vs RS67)**

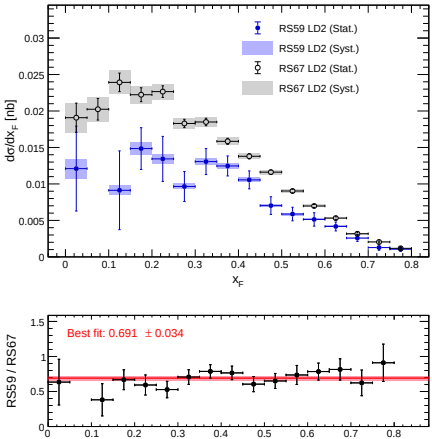


# Cross-Section Ratio RS59 to RS67 (xF bins)

**LH2 (RS59 vs RS67)**

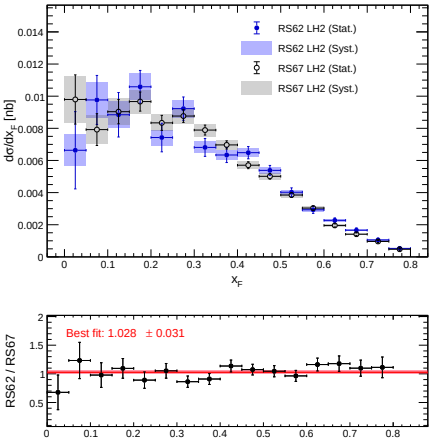



**LD2 (RS59 vs RS67)**

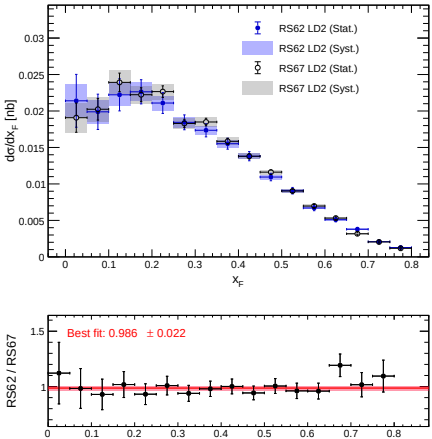


# Cross-Section Ratio RS62 to RS67 (xF bins)

**LH2 (RS62 vs RS67)**

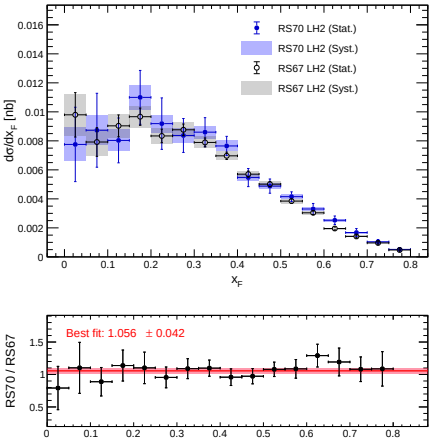



**LD2 (RS62 vs RS67)**

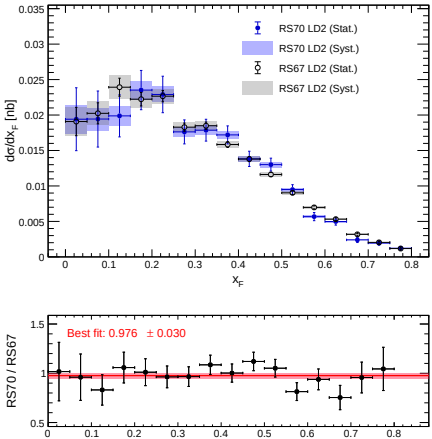


# Cross-Section Ratio RS70 to RS67 (xF bins)

**LH2 (RS70 vs RS67)**

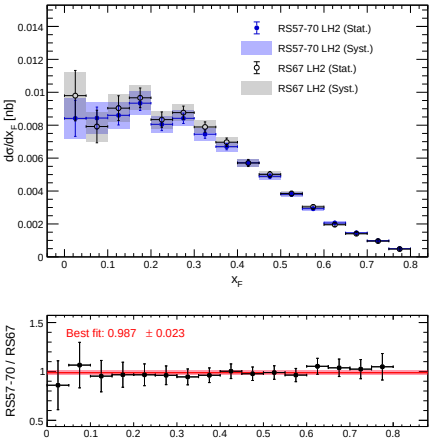



**LD2 (RS70 vs RS67)**

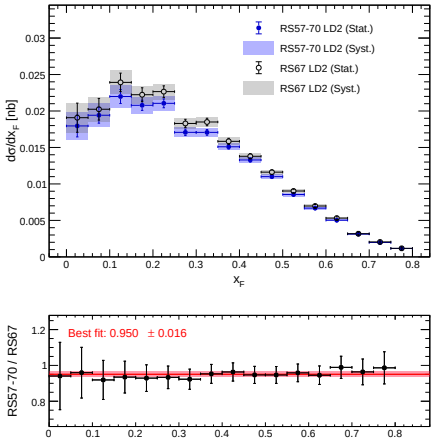


# Cross-Section Ratio RS57-70 to RS67 (xF bins)

**LH2 (RS57-70 vs RS67)**




**LD2 (RS57-70 vs RS67)**



# Cross-Section comparison to RS67

Table 1: Best-fit single differential cross-section ratios vs. pT (RS/RS67) for $LH_2$ and $LD_2$ targets.

| Roadset Ratio | $LH_2$ Best Fit ($\pm$ Error) | $LD_2$ Best Fit ($\pm$ Error) |
|---|---|---|
| RS57 / RS67 | $0.773 \pm 0.036$ | $0.736 \pm 0.022$ |
| RS59 / RS67 | $0.748 \pm 0.049$ | $0.683 \pm 0.034$ |
| RS62 / RS67 | $1.020 \pm 0.035$ | $0.981 \pm 0.023$ |
| RS70 / RS67 | $1.053 \pm 0.046$ | $0.991 \pm 0.031$ |
| RS57-70 / RS67 | $0.976 \pm 0.027$ | $0.945 \pm 0.018$ |

Table 2: Best-fit single differential cross-section ratios vs. xF (RS/RS67) for $LH_2$ and $LD_2$ targets.

| Roadset Ratio | $LH_2$ Best Fit ($\pm$ Error) | $LD_2$ Best Fit ($\pm$ Error) |
|---|---|---|
| RS57 / RS67 | $0.804 \pm 0.034$ | $0.745 \pm 0.021$ |
| RS59 / RS67 | $0.820 \pm 0.050$ | $0.691 \pm 0.034$ |
| RS62 / RS67 | $1.028 \pm 0.031$ | $0.986 \pm 0.022$ |
| RS70 / RS67 | $1.056 \pm 0.042$ | $0.976 \pm 0.030$ |
| RS57-70 / RS67 | $0.987 \pm 0.023$ | $0.950 \pm 0.016$ |

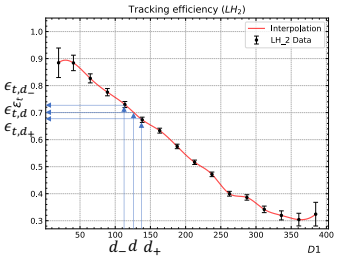
# Next Steps

- During the meeting Paul Reimer pointed out that the possibility of using a different version of kTracker between runs 2 and 3. Is it possible to check the difference between run 2 and 3 reconstructions?

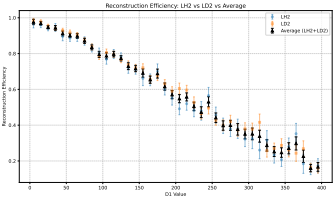
# Backup Slides

# New Reconstruction Efficiency Correction (D1-10)

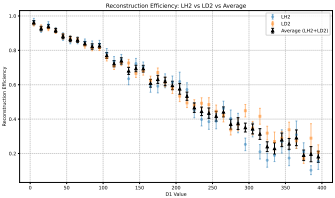
**Previous Global Curve (All RS)**



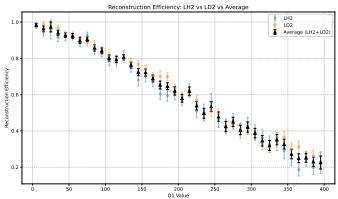

**RS57 Efficiency Curve**



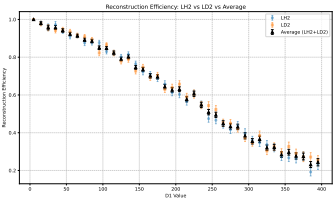

**RS59 Efficiency Curve**



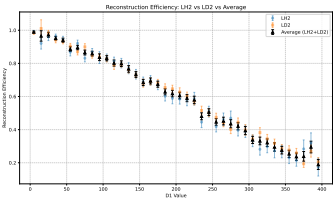

**RS62 Efficiency Curve**




**RS67 Efficiency Curve**

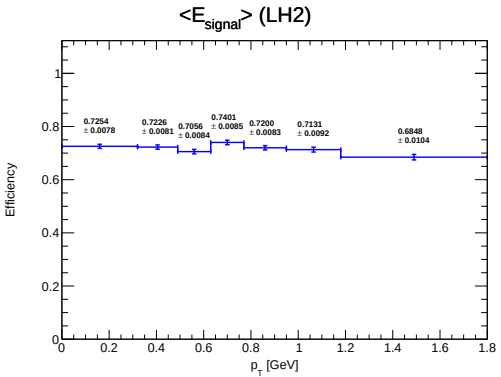



**RS70 Efficiency Curve**

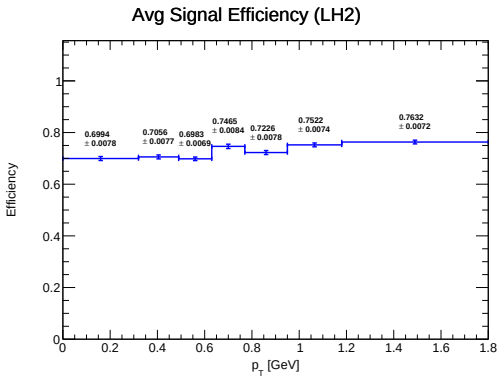


# RS57: Latest Vs Existing Reconstruction Efficiencies (LH2)

**Latest Efficiency (RS57 LH2)**

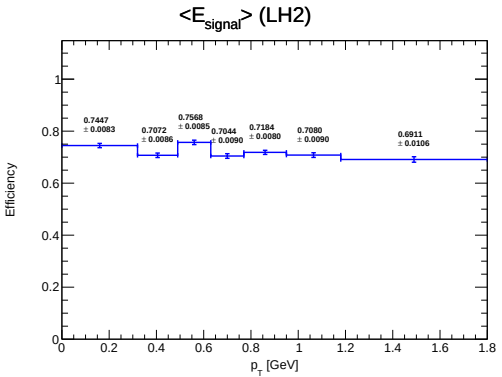



**Existing Efficiency (RS57 LH2)**

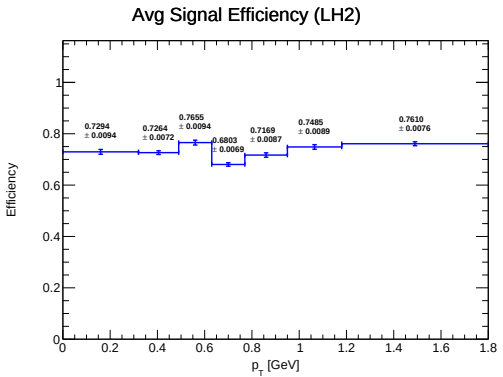


# RS59: Latest Vs Existing Reconstruction Efficiencies (LH2)

**Latest Efficiency (RS59 LH2)**

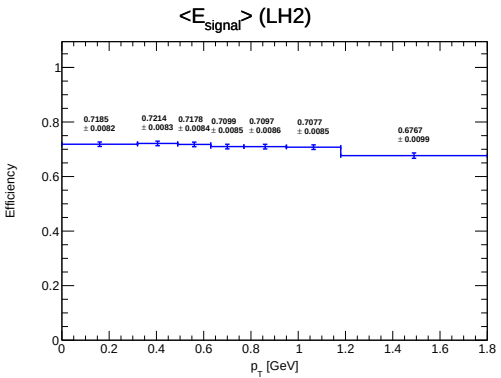



**Existing Efficiency (RS59 LH2)**

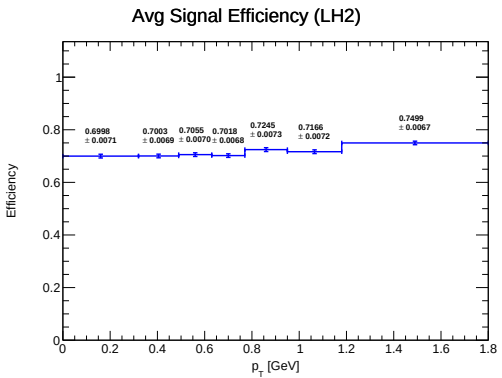


# RS62: Latest Vs Existing Reconstruction Efficiencies (LH2)

**Latest Efficiency (RS62 LH2)**

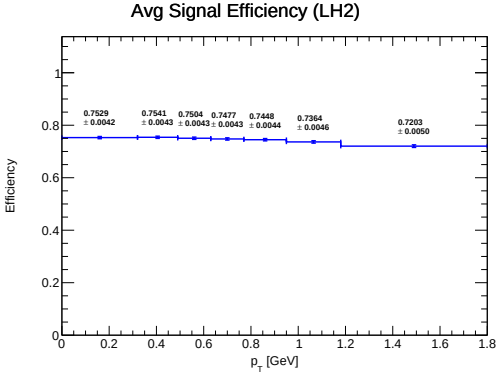



**Existing Efficiency (RS62 LH2)**

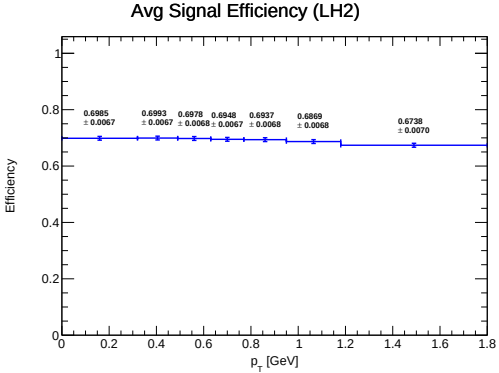


# RS67: Latest Vs Existing Reconstruction Efficiencies (LH2)

**Latest Efficiency (RS67 LH2)**

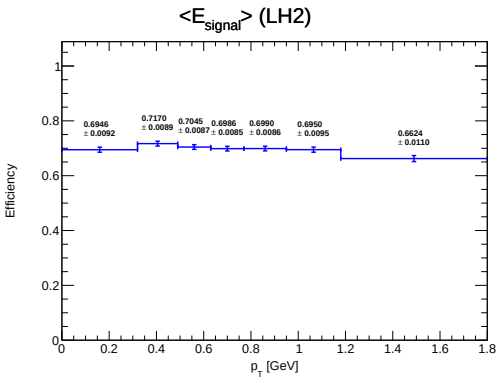



**Existing Efficiency (RS67 LH2)**

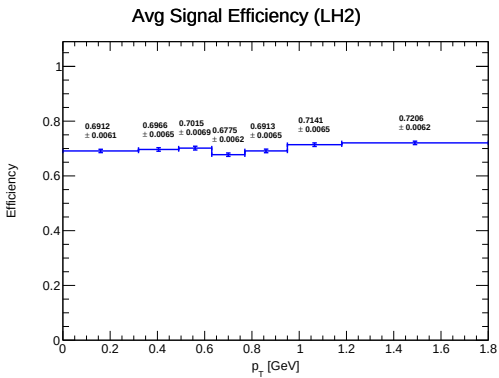


# RS70: Latest Vs Existing Reconstruction Efficiencies (LH2)

**Latest Efficiency (RS70 LH2)**

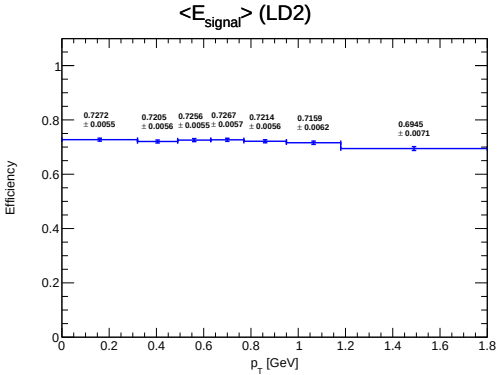



**Existing Efficiency (RS70 LH2)**

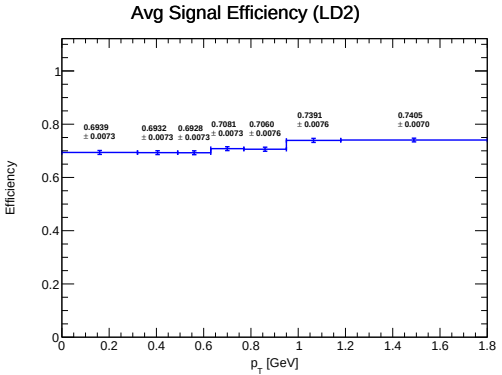


# RS57: Latest Vs Existing Reconstruction Efficiencies (LD2)

**Latest Efficiency (RS57 LD2)**

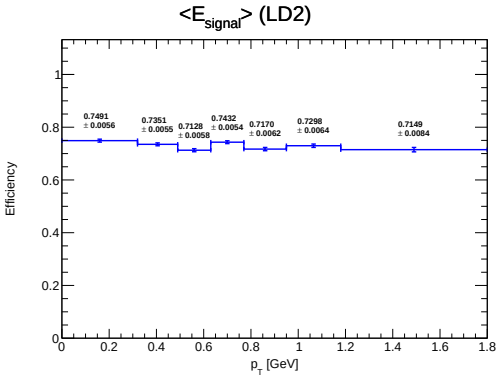



**Existing Efficiency (RS57 LD2)**

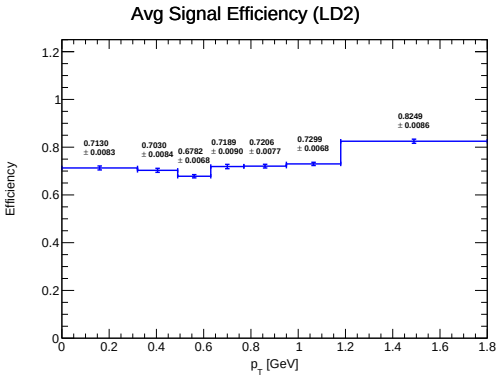


# RS59: Latest Vs Existing Reconstruction Efficiencies (LD2)

**Latest Efficiency (RS59 LD2)**

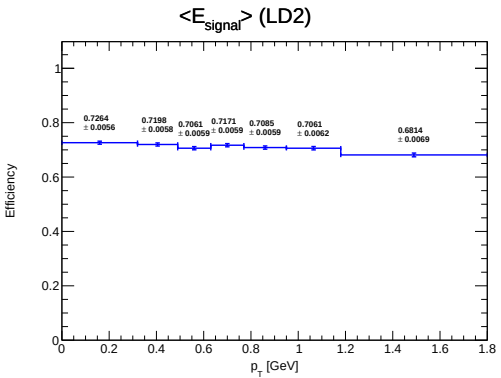



**Existing Efficiency (RS59 LD2)**

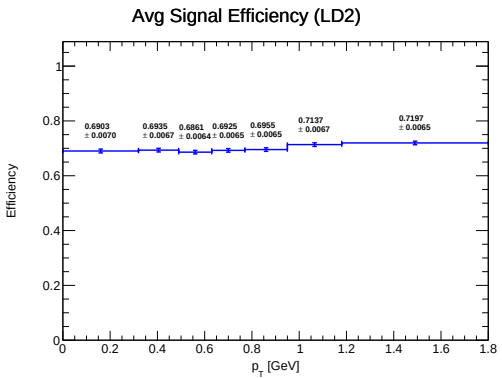


# RS62: Latest Vs Existing Reconstruction Efficiencies (LD2)

**Latest Efficiency (RS62 LD2)**

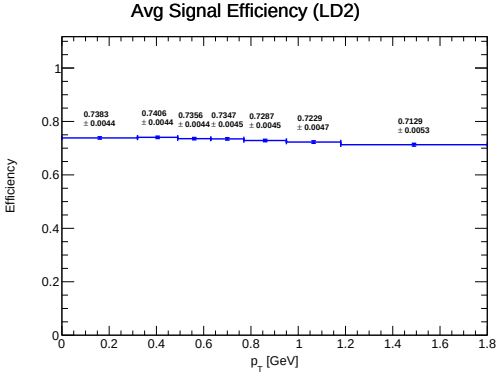



**Existing Efficiency (RS62 LD2)**

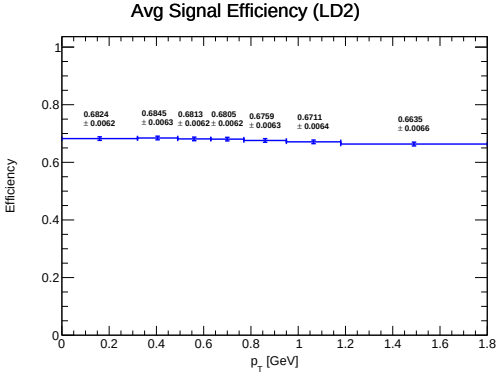


# RS67: Latest Vs Existing Reconstruction Efficiencies (LD2)

**Latest Efficiency (RS67 LD2)**

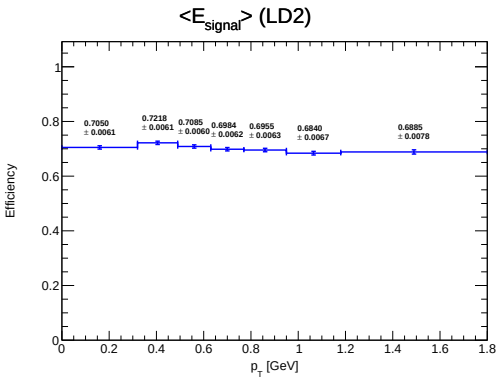



**Existing Efficiency (RS67 LD2)**

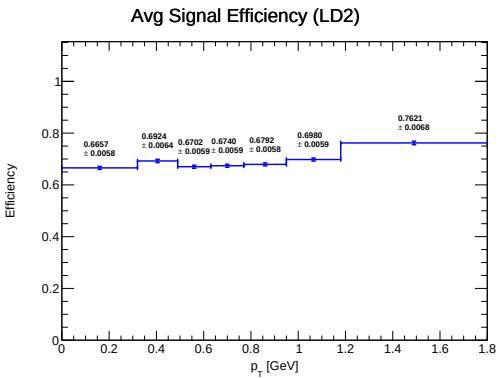


# RS70: Latest Vs Existing Reconstruction Efficiencies (LD2)

**Latest Efficiency (RS70 LD2)**

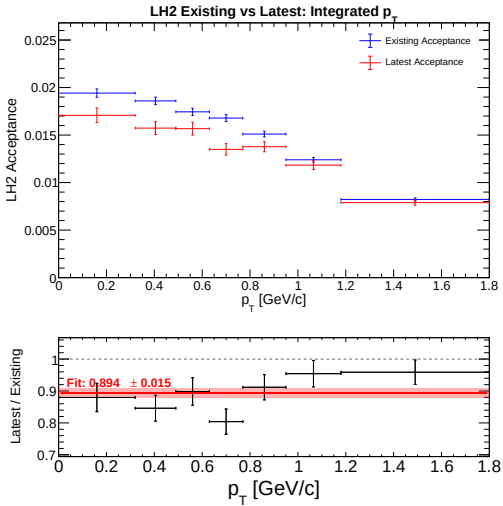



**Existing Efficiency (RS70 LD2)**

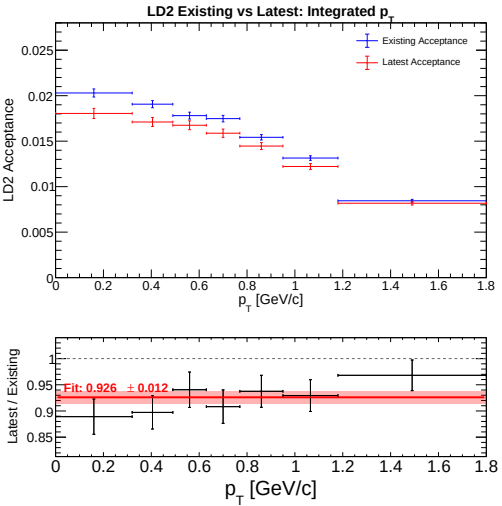


# RS57 Acceptance Corrections

**LH2 (RS57)**

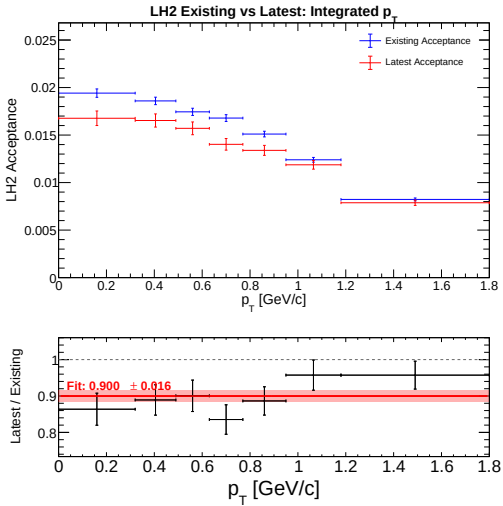



**LD2 (RS57)**

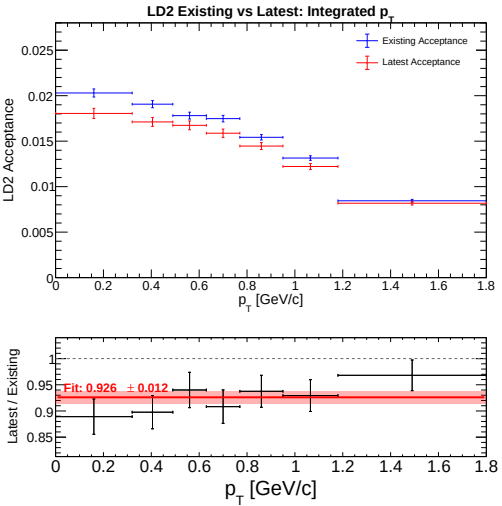


# RS59 Acceptance Corrections

**LH2 (RS59)**

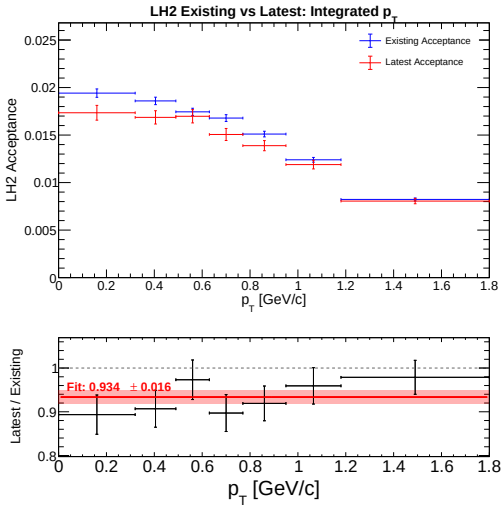



**LD2 (RS59)**

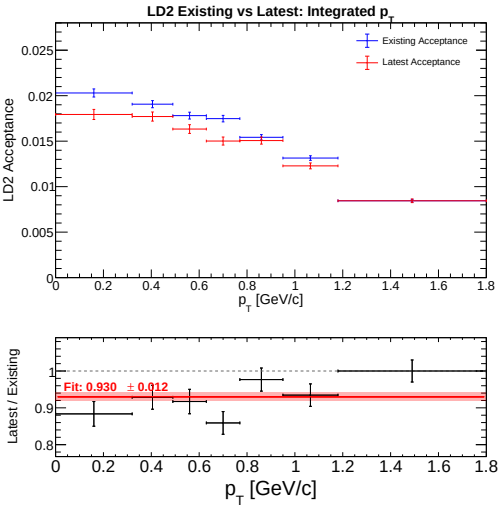


# RS62 Acceptance Corrections

**LH2 (RS62)**

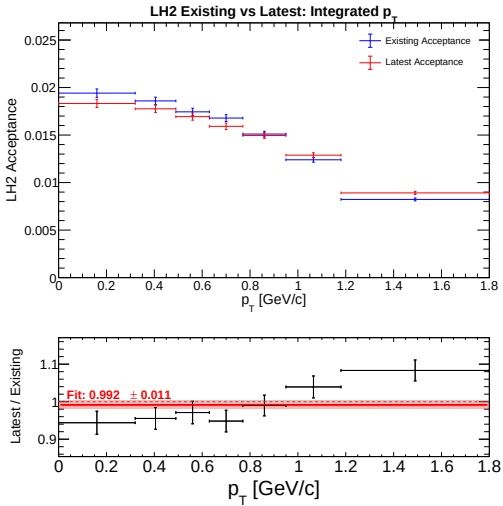



**LD2 (RS62)**

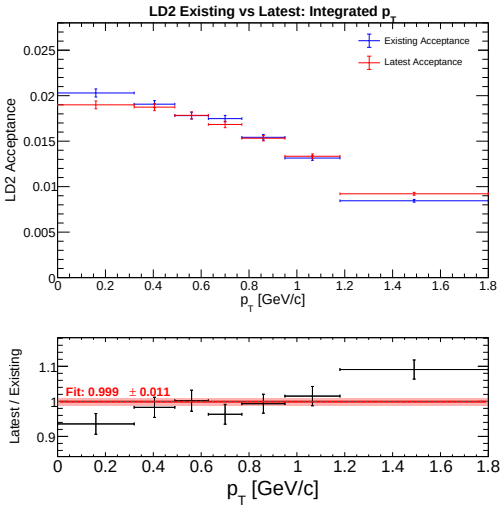


# RS67 Acceptance Corrections

**LH2 (RS67)**

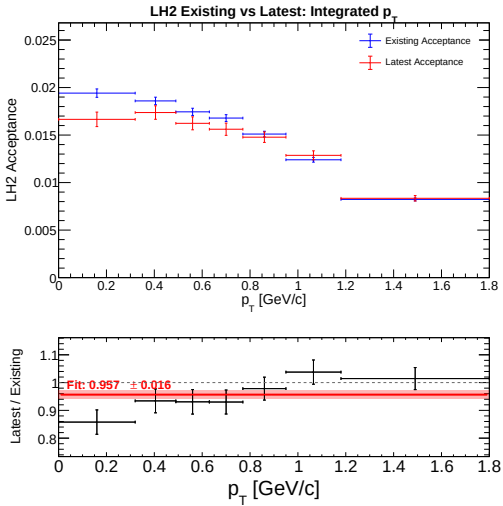



**LD2 (RS67)**

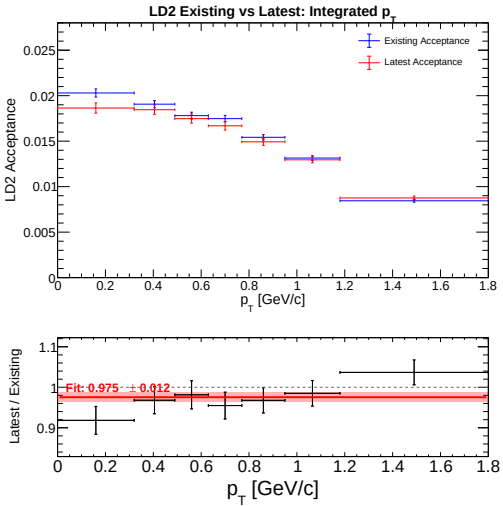


# RS70 Acceptance Corrections

**LH2 (RS70)**

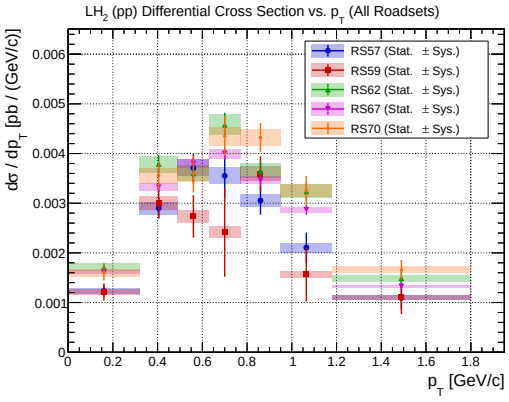



**LD2 (RS70)**

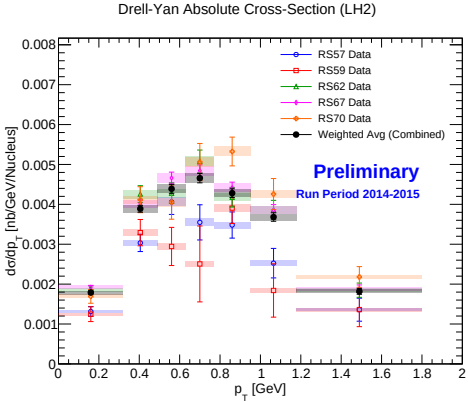


# LH2 Cross-Section Comparison All Roadsets

**Latest (LH2 All Roadsets)**

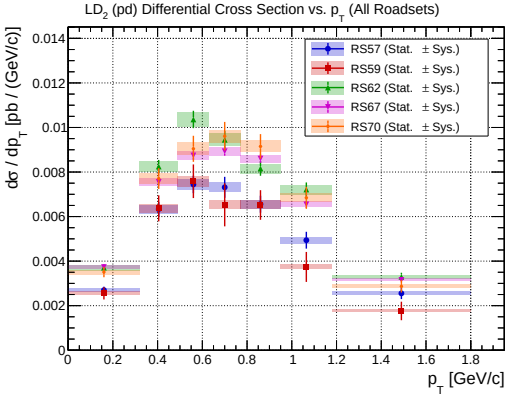



**Existing (LH2 All Roadsets)**



# LD2 Cross-Section Comparison All Roadsets

**Latest (LD2 All Roadsets)**




**Existing (LD2 All Roadsets)**

